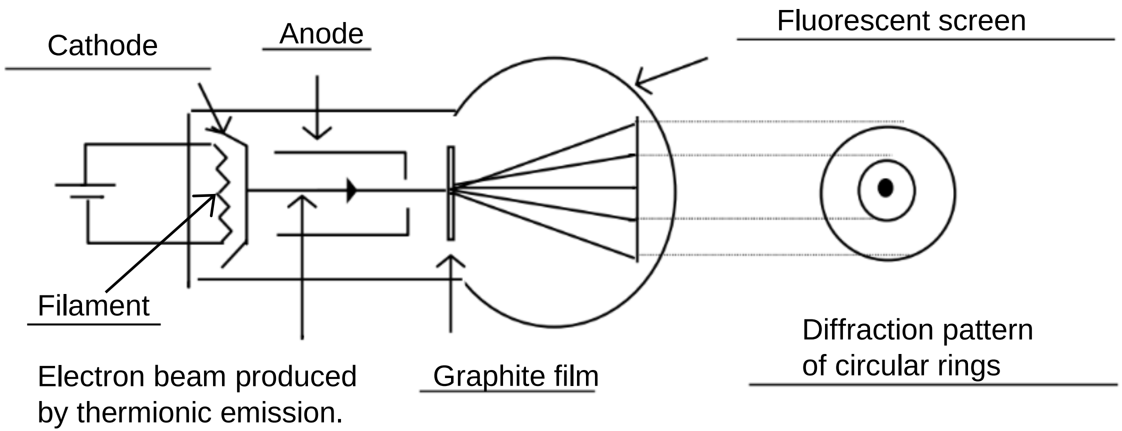
Q6(a)



[2]

[1 mark for correct diagram and1 mark for suitable labelling. Allow for 1 missing label. Cathode is optional]

- Electrons boil off the heated filament by a process called thermionic emission. The electrons are accelerated towards the anode. [1]

- The electrons then impinge upon a thin graphite film and undergo diffraction as waves when they pass through the carbon film. [1]

- When these diffracted electrons impinge upon the fluorescent screen, a diffraction pattern of circular rings is observed. [1]

Q6(b)

From conservation of energy,

Loss in electric potential energy of electron = gain in kinetic energy of electron [1]

$$q\Delta V = \frac{p^2}{2m}$$

$$(1.6\times10^{-19})(5.0\times10^{3}) = \frac{p^2}{2(9.11\times10^{-31})}$$ [1]

$p = 3.8179\times10^{-23} = \mathbf{3.82\times10^{-23}\ kg\ m\ s^{-1}}$ [1]

Q6(c)

The de Broglie wavelength equation states that $\lambda = \dfrac{h}{p}$ [1]

$\lambda = \dfrac{6.63\times10^{-34}}{3.8179\times10^{-23}} = 1.737\times10^{-11} = \mathbf{1.74\times10^{-11}\ m}$ [1]

Q7(a)(i)